Systematic Value in Developed Equities

Investment note — design, five-year backtest (June 2021 – June 2026) and current portfolio

Noah Doose · MSc Financial Markets & Investments, SKEMA Business School · October 2026. Student research prepared for interview purposes; not investment advice. Data: Bloomberg (SKEMA terminal, extracted 26-29/10/2026), Kenneth French Data Library, OECD/FRED. No security-level raw data is reproduced.

1. Executive summary

- **Mandate.** Personal portfolio, 80% equities / 20% cash. The equity sleeve runs a systematic value strategy on developed-market large and mid caps, benchmarked against MSCI World Net Total Return in EUR.
- **Process.** Four valuation yields (EV/EBIT, P/B, FCF yield, EV/Sales; banks and insurers: P/B and P/E), winsorised and z-scored within GICS sectors, with a strict accounting lag (fiscal year-end + 6 months). Fifty names are chosen from the cheapest sector quintile under ±10-point sector and country bands, then weighted equally, by minimum variance or by mean-variance.
- **Result.** Equal weight returned 11.6% a year net of costs and withholding tax, hedged per the IPS, versus 11.7% for the benchmark (active -0.1%, tracking error 10.8%). Unhedged it returned 12.7% (+0.9%). Minimum variance and mean-variance lagged by about 2.3 points a year.
- **Why.** Stock selection inside sectors added +2.3% a year, sector allocation cost -1.4% (no mega-cap growth), and the USD hedge -0.8% (rate differential). The active return loads on HML (0.53, t = 6.4) and negatively on momentum (-0.27); no residual alpha.
- **Signal.** The score's rank information coefficient against next-six-month sector-relative returns averaged 0.086 (t = 2.1); the cheapest sector-neutral quintile beat the dearest by 8.1% a year.
- **Integrity.** The universe is rebuilt at each date from the dated index composition (no survivorship), accounting data are lagged and never used before publication, and the backtest caught and corrected a dividend double count worth about 2% a year (Bloomberg prices were already dividend-adjusted).

Table 1. Equity sleeve, 2 Jul 2021 – 26 Jun 2026, EUR, hedged per IPS, net of 15 bp costs and withholding tax

	Ann. return	Active	Volatility	Sharpe	TE	IR	Max DD	Turnover/yr
Equal weight	11.6%	-0.1%	15.8%	0.61	10.8%	0.02	-17.8%	57%
Minimum variance	9.4%	-2.3%	13.6%	0.55	10.5%	-0.20	-19.1%	70%
Mean-variance	9.4%	-2.4%	15.7%	0.48	10.4%	-0.18	-21.8%	70%
MSCI World Net EUR	11.7%	—	13.7%	0.72	—	—	-18.2%	—

Source: outputs/backtest/performance_summary.csv

2. Client and investment policy

The client is the author: 21 years old, long horizon, human capital largely uncorrelated with equity markets, low liquidity needs. Risk capacity is above average; the strategic allocation is 80% equities and 20% cash, rebalanced with the equity strategy. The equity objective is to beat MSCI World Net (EUR) after costs through a value premium rather than market timing, with a process that can be defended line by line.

Table 2. IPS summary (locked decisions)

Parameter	Rule
Universe	Bloomberg World Large & Mid Cap, 24 developed countries, composition dated at each rebalance
Benchmark	MSCI World Net Total Return EUR (MSDEWIN)
Signal	Equal-weight z-scores of EV/EBIT, P/B, FCF yield, EV/Sales (financials: P/B, P/E), winsorised 1/99, within

	GICS sector
Selection	Top quintile of the score within each sector
Construction	40-50 names, 0.5%-4% each, sector and country within ±10 pts of benchmark, long only, >98% invested
Rebalancing	Semi-annual (June/December), 10 dates from 30/06/2021 to 31/12/2025
Accounting lag	Fiscal year usable from year-end + 6 months (robustness: announcement + 3 months)
Liquidity	Free-float cap > EUR 1bn; 3-month ADV > EUR 5m
Costs	15 bp per trade; target turnover < 60% a year
Currency	EUR reporting; a currency carried by 5 lines or more is hedged with forwards

Source: docs/IPS_Portefeuille_Personnel.md, CLAUDE.md

3. Data and universe

All market and accounting data were extracted from the Bloomberg terminal into a single, immutable workbook and transformed by script. The investable universe at each date is the index composition at that date, restricted to developed countries by country of risk, filtered for liquidity and reduced to one line per issuer. It shrinks from about 1,640 to 1,100 names as the Bloomberg index has fewer members, and always covers about 98% of the developed-market index weight.

Dividends counted twice, caught. The key data finding concerns dividends. The plan was to add dividends to price returns. A replica of the benchmark built that way beat MSCI World Net by two points a year, and even a price-only replica beat it. An event study over roughly 38,000 ex-dividend weeks showed that prices do not drop on ex-dates: the extracted prices were already adjusted for dividends. Total return is therefore the adjusted price return, and withholding tax is deducted to match the net benchmark. After the fix the replica tracks MSCI World Net with 0.996 weekly correlation and a 0.3-point annual gap.

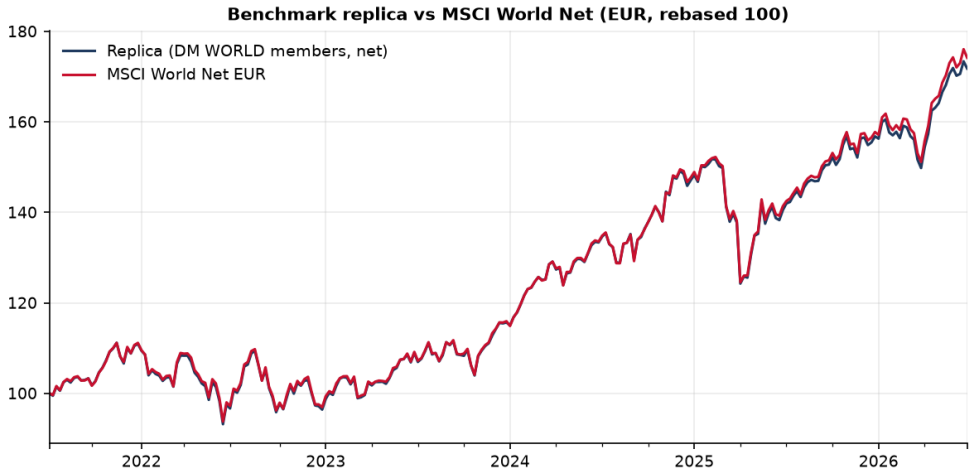


Figure 1. Benchmark replica from security returns vs MSCI World Net EUR

Source: outputs/data/benchmark_replica.png; docs/data_quality.md

4. The value signal

For each company the latest fiscal year that was public at the decision date is used. Enterprise value is the market capitalisation at the date plus net debt and minorities. The four yields are winsorised at the 1st and 99th percentiles across the universe, standardised within GICS sectors and averaged; banks and insurers use book-to-price and earnings yield only. Ranking within sectors turns the signal into a stock-picking tool and leaves sector exposure to the risk bands.

The signal carried information over the period, before any portfolio construction: mean rank IC 0.086 (positive in 7 of 10 periods), a 8.1% annual spread between the cheapest and dearest sector-neutral quintiles, and a stable ranking (rank correlation 0.90 from one date to the next). The losing periods — first half of 2023, 2024 — coincide with the concentrated rally in AI-related growth stocks.

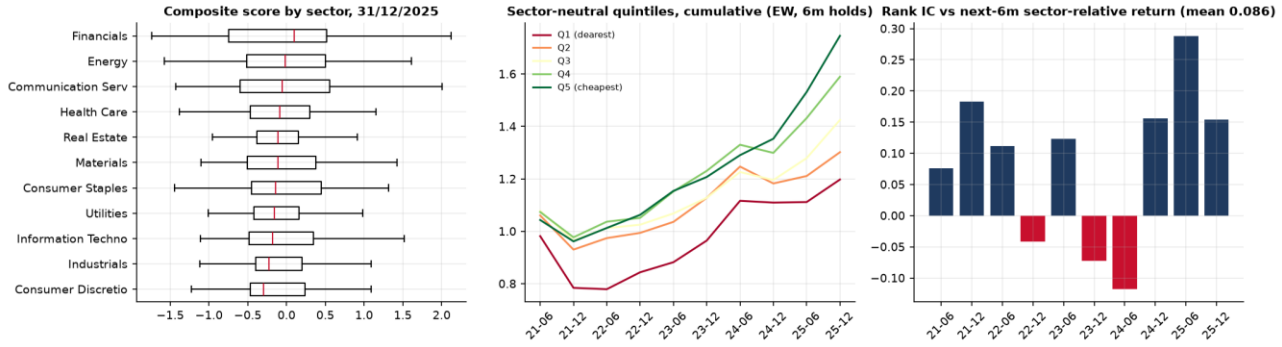


Figure 2. Score by sector, quintile returns and information coefficient

Source: outputs/scores/

5. Portfolio construction

Fifty names are selected once per date by an exact integer programme that maximises the sum of scores under the sector and country bands; the three methods then weight the same names, which isolates the value of the weighting scheme. Equal weight is the reference (DeMiguel, Garlappi and Uppal, 2009). Minimum variance uses a Ledoit-Wolf shrunk covariance on 250 weekly returns; mean-variance adds historical mean returns as a deliberately naive robustness test. A turnover buffer keeps holdings that remain in the top two quintiles; it was set to meet the turnover target, not to maximise performance. All 30 portfolios satisfy every IPS constraint (outputs/portfolio/constraint_check.csv).

At 31 December 2025 the equal-weight portfolio is underweight the US by about 9 points and information technology by about 9 points, overweight France, Germany and Japan, and overweight health care, industrials, consumer discretionary and financials. Its top-10 weight is 20%; minimum variance concentrates more (top-10 about 40%).

6. Performance

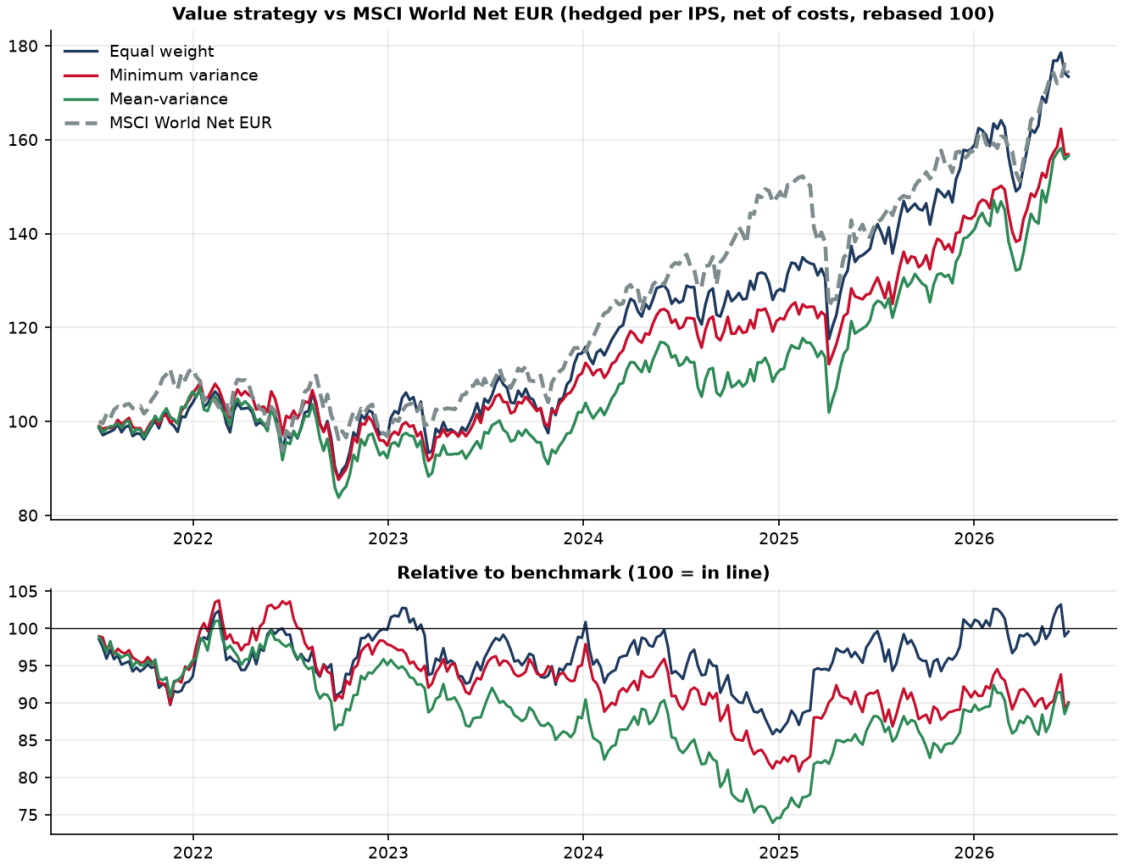


Figure 3. Cumulative performance and relative performance vs benchmark

Source: outputs/backtest/nav_weekly.csv

Table 3. Calendar-year returns, EUR

	Equal weight	Minimum variance	Mean-variance	MSCI World Net EUR
2021 (H2)	4.0%	5.9%	6.4%	11.1%
2022	-6.9%	-10.4%	-13.3%	-12.8%
2023	18.3%	16.2%	10.6%	19.6%
2024	11.5%	10.1%	8.0%	27.4%
2025	23.8%	17.9%	27.2%	6.8%
2026 (H1)	9.7%	9.6%	11.7%	10.4%

Source: outputs/backtest/calendar_returns.csv

Equal weight roughly matched the benchmark over five years, with large swings in relative performance: it lagged by 16 points in 2024, when the market was led by a handful of mega-cap technology names it could not own, and led by 17 points in 2025. Costs took about 0.2% a year. Hedging cost about 1% a year because a euro investor hedging dollars pays the interest-rate differential, and the dollar did not fall enough to compensate. For the total 80/20 portfolio, equal weight returned 9.9% against 9.9% for the 80/20 benchmark.

7. Risk and attribution

Table 4. Brinson-Fachler attribution vs benchmark replica, % a year (arithmetic)

	Sector allocation	Selection within sectors	FX overlay

Equal weight	-1.35%	+2.27%	-0.77%
Minimum variance	-1.54%	-0.18%	-0.25%
Mean-variance	-1.45%	+0.48%	-1.01%

Source: outputs/risk/attribution_summary.csv

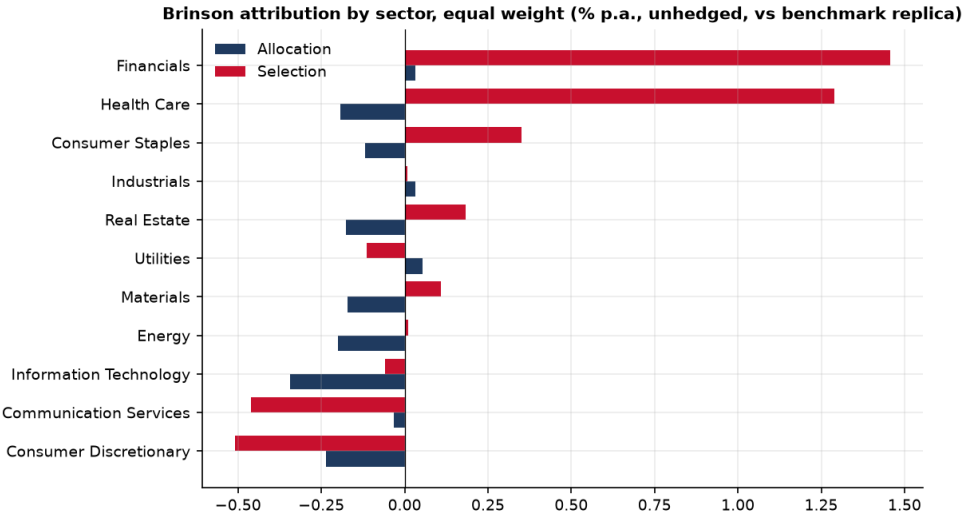


Figure 4. Attribution by sector, equal weight

Source: outputs/risk/brinson_gics_EW_total.csv

Table 5. Factor loadings of weekly active returns (Fama-French Developed + momentum, Newey-West)

	Equal weight	Minimum variance	Mean-variance
Alpha (annualised)	-0.6%	-1.7%	-2.4%
Market (EUR)	0.08	-0.10	0.06
Size (SMB)	0.33	0.22	0.30
Value (HML)	0.53	0.33	0.31
Profitability (RMW)	0.00	0.12	0.10
Investment (CMA)	-0.00	0.23	0.20
Momentum	-0.27	-0.26	-0.17
R^2	0.58	0.56	0.35

Source: outputs/risk/factor_regressions.csv; Kenneth French Data Library (downloaded 29/10/2026)

The decomposition confirms the portfolio is what it claims to be: a value tilt (HML strongly significant), with a small-size and an anti-momentum tilt, and no statistically meaningful alpha. Selection added value where valuations were most dispersed (European and Japanese financials, US managed care); allocation and the hedge took it back. Minimum variance adds a defensive quality and investment tilt and a lower beta, which this period did not reward.

8. Robustness

Table 6. Robustness: each variant changes one assumption

Variant	Method	Ann. return	Active	TE	Turnover/yr
Base (FYE+6m, quintile, buffer)	Equal weight	11.6%	-0.1%	10.8%	57%
Base (FYE+6m, quintile, buffer)	Minimum variance	9.4%	-2.3%	10.5%	70%
Lag: announcement + 3 months	Equal weight	11.5%	-0.3%	10.8%	58%
Lag: announcement + 3 months	Minimum variance	10.3%	-1.4%	10.6%	70%
Pool: top tercile, largest caps (not best scores)	Equal weight	11.1%	-0.6%	8.9%	27%
Pool: top tercile, largest caps (not best scores)	Minimum variance	9.3%	-2.5%	10.0%	30%
Pool: quintile, no turnover buffer	Equal weight	11.4%	-0.3%	11.0%	83%
Pool: quintile, no turnover buffer	Minimum variance	9.1%	-2.6%	10.8%	97%
Excluding financials	Equal weight	10.3%	-1.5%	10.3%	52%

Excluding financials	Minimum variance	10.1%	-1.6%	10.5%	65%
Gross dividends (no withholding)	Equal weight	12.5%	+0.8%	10.8%	57%
Gross dividends (no withholding)	Minimum variance	10.2%	-1.6%	10.5%	70%
Weights proportional to the score (base selection)	Score-tilted	11.5%	-0.2%	11.1%	67%

Source: outputs/backtest/robustness.csv

No variant produces a robust outperformance, and none changes the conclusion: the lag convention, the turnover buffer and the choice of the most extreme scores move results by less than half a point; dropping financials costs 1.5 points, because selection among banks and insurers was the largest contributor. Gross dividends would flatter results by about 0.9 points a year, which is why the headline uses net returns.

9. Current portfolio and stock pitches

Eight holdings of the 31/12/2025 portfolio were analysed fundamentally, each in a separate deck: six conviction cases and two deliberate cautionary cases where the screen is likely wrong.

Table 7. The eight pitches (decks in pitches/)

Company	Call	Thesis
Renault	Buy	Net cash plus the finance arm cover most of the price; car business, Nissan stake and product cycle come almost free.
BNP Paribas	Overweight	13% RoTE at 0.8x tangible book and ~7% yield: the French sovereign discount has gone too far.
Carrefour	Buy	10%+ free-cash-flow yield after a profit warning; real estate covers the enterprise value.
Centene	Neutral (case study)	Cautionary case: cheap on stale 2024 earnings; the 2026 recovery was not what the signal forecast.
HP Inc	Overweight	~10x earnings and ~10% FCF yield while the memory-cost squeeze is temporary.
Nippon Steel	Buy	Overseas profit about to double (U.S. Steel, India) at 0.5-0.6x book.
Orient Overseas (International)	Underweight (case study)	Cautionary case: the highest score in the universe, on cyclical peak earnings.
Kansai Electric Power	Hold	Japan's nuclear leader, modestly cheap once record 2025 profits normalise.

Source: pitches/.pptx; pitch sources slide in each deck*

10. Limitations and next steps

- **Short sample.** Five years and ten rebalances: an information ratio is estimated with a standard error of about 0.45. The results describe one regime (rate shock, then an AI-led growth rally).
- **Universe vs benchmark.** Bloomberg's index stands in for MSCI's (constituents not licensed). The replica test bounds the mismatch (0.996 correlation).
- **Classification.** Current GICS sectors are applied to the whole history.
- **Dividend currency.** The extract lacks the payment currency; it is inferred for UK and Israeli lines and only affects the withholding haircut.
- **Costs and hedging.** No market impact; forward points from 3-month interbank rates, re-struck weekly.
- **What I would do next.** Combine value with momentum and quality (the anti-momentum loading is the main drag), test a cap-aware weighting to cut tracking error, and compare the hedge policy against a hedged benchmark.

Reproducibility

python -m src.run_all rebuilds every number and figure from the raw workbook; tests in tests/ check the accounting lag, total returns and every portfolio constraint. Decisions and their alternatives are logged in docs/decisions.md; the bias checklist is in docs/bias_checklist.md.

Appendix A — Risk detail

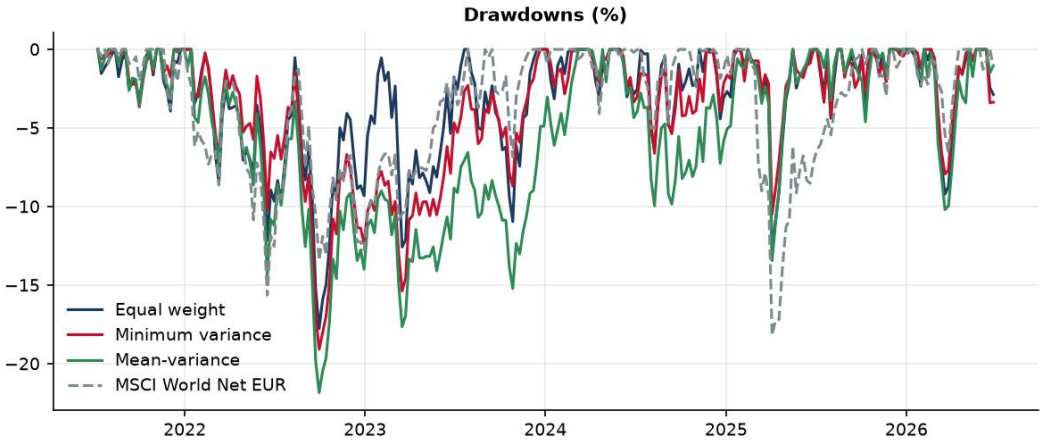


Figure 5. Drawdowns of the three methods and the benchmark

Source: outputs/backtest/nav_weekly.csv

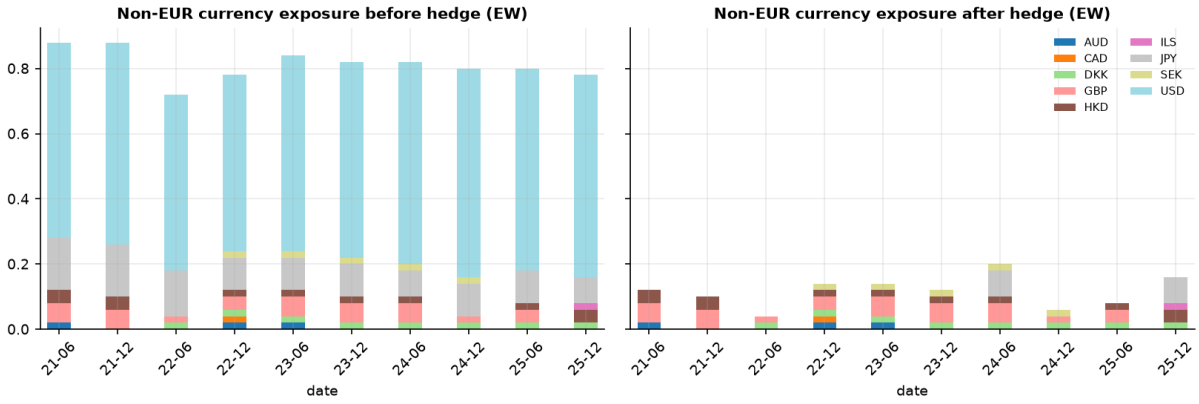


Figure 6. Non-euro currency exposure before and after the IPS hedge (equal weight)

Source: outputs/backtest/fx_exposure_before_after_hedge.csv

Table 8. Ex-ante risk contribution by sector at 31/12/2025, equal weight (portfolio volatility 13.8%)

	Weight	Share of ex-ante risk
Communication Services	2%	1%
Consumer Discretionary	14%	18%
Consumer Staples	6%	3%
Financials	20%	20%
Health Care	16%	8%
Industrials	16%	17%
Information Technology	18%	25%
Materials	4%	4%
Real Estate	2%	3%
Utilities	2%	1%

Source: outputs/risk/risk_contribution_by_sector_last_date.csv

Appendix B — Data integrity

Table 9. Event study: weekly price return on the ex-date regressed on the dividend yield (unadjusted prices would give a slope near −1)

Dividend yield of the week	Events	Mean yield	Slope on yield	t-stat
0%-2%	32164	0.008	-0.049	-1.209
2%-5%	5781	0.030	-0.036	-0.555
5%-25%	834	0.068	0.116	1.361

Source: docs/data_quality.md §3

Other controls: market capitalisation vs price × shares (median ratio 1.07), no weekly price move above ±50%, dividend currency inferred for 28 UK and Israeli lines, 471 fiscal years published more than six months after year-end and therefore blocked until publication, free float above 100% capped (21 readings).

Appendix C — Anti-bias checklist

Table 10. Checklist written before interpreting the backtest

Risk	Control	Status
Survivorship	Universe at each date = WORLD composition at that date (10 dated lists), never today's members nor the union	OK
Survivorship (holding period)	A held line that stops trading earns 0% after its last price; no forward-filled gains	OK — limit: a bankruptcy with no final price would be missed (none observed in holdings)
Look-ahead (accounting)	Fiscal year usable from FYE + 6 months and not before announcement	OK
Look-ahead (restatements)	Originally reported figures (`FILING_STATUS = OR`) at extraction	OK (to confirm at terminal)
Look-ahead (prices / trading)	Decisions use information up to D (cap/float/turnover snapshots at D, returns up to the last Friday ≤ D); trades at the close of the first Friday after D	OK
Look-ahead (classification)	GICS sector = current classification applied to the whole history	Open — limited impact (few re-classifications, mainly 2023 GICS changes for payments/Visa/Mastercard moving to Financials)
Units: dividends double-counted	Event study proves `PX_LAST` already embeds dividends: TR = adjusted price return	Fixed (was +2%/yr)
Units: pence / sub-units	Prices in EUR at source; dividend currency inferred for GBp/ILs (only used for the withholding haircut)	OK
Units: millions / EUR	Fundamentals and market cap both in EUR millions; ratios of company totals only	OK
Gross vs net dividends	Headline uses net-of-withholding returns (MSCI net convention); gross reported as sensitivity (+0.9%/yr for EW)	OK
Index provider mismatch	Universe Bloomberg World DM vs benchmark MSCI World	Disclosed
Transaction costs	15 bps × two-sided turnover, initial build included; no market impact	Disclosed (impact understated for less liquid names)
FX hedge cost	Carry from 3-month interbank differentials (covered interest parity), weekly re-struck hedge	Disclosed (no forward bid/ask, weekly re-strike idealised)
Data snooping in construction	Turnover buffer fixed before seeing the corrected results (target < 60%/yr); a larger buffer did better but was not adopted	OK
Small sample	5 years, 10 rebalances, one regime	Disclosed

Source: docs/bias_checklist.md

Weights proportional to the score were also tested on the same selection: 11.5% a year, -0.2% vs the benchmark — equal weight is kept as the reference.